Closing the Loop: Preference Geometry and LLM Judgment as a Single System

On combining decorrelated preference tuning (Blair, Procaccia & Tambe) with the ELJ claim registry

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Companion: `REPORT-1-ELJ-ENGINE.md`
Subject paper: Carter Blair, Ariel D. Procaccia, Milind Tambe, *Embeddings for Preferences, Not Semantics.*

Abstract

Two independent lines of work reach the same diagnosis — off-the-shelf sentence embeddings encode topic and wording, not stance — and adopt opposite remedies. Blair et al. **repair the geometry**, fine-tuning an encoder on synthetic hard triplets (48.3% → 80.0% triplet accuracy) or learning a per-topic rank-20 projection from participant votes (81.1%). We **bypass the geometry**, demoting cosine to a candidate ranker and putting an LLM judge on every decision (95.0% in isolation, 75.3% in a live corpus replay).

This report argues the two are not competitors but the two halves of one system, and identifies a **closed loop** neither has alone:

- Their tuned geometry solves our dominant remaining failure — candidate lists ordered adversarially, with distractors outranking true matches **98.1%** of the time.
- Our claim registry solves their stated open problem — obtaining topic-specific supervision without per-topic votes. Their paper names this as future work; our engine already emits it as a by-product of normal operation.

We specify four hybrid architectures, give quantitative predictions with the evidence behind each, and set out the experiments that would test them. We also identify a shared blind spot: **both methods have been validated only against adversarial examples engineered to defeat embeddings**, and a combined system requires an adversary built against LLM judgment.

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1. One diagnosis, two remedies

1.1 The agreement is exact

	Blair et al.	This work
cosine cannot see stance	base encoders 6.3–48.3% on hard triplets	`text-embedding-3-small` : **0.5%**
distractors beat matches	"rank the semantic distractor above the preference match 70–95% of the time"	**863/880 = 98.1%**
natural-data signal is illusory	50pp drop, natural → hard	99.1% of contradictions score ≥ 0.85

Their §4.1 supplies the theory for what we measured empirically. The cosine margin decomposes as

> s(a,p) − s(a,n) = Δ_S + Δ_T

where Δ_S lies in the stance-relevant subspace and Δ_T is nuisance (wording, style, topic). Cosine weights both equally. On natural deliberation data the two correlate; hard triplets break the correlation and cosine collapses.

This explains our operational finding precisely: **0.85 is safe as a fetch floor and catastrophic as a decision threshold**, because the 0.85 is mostly Δ_T. Their formalism is the "why" behind our Table 2.

1.2 The remedies are orthogonal

```
statement pair
     │
     ├─ EMBED ───► THEIR FIX: change the geometry
     │             DPT fine-tune (§5) or per-topic rank-20 projection (§7)
     │             → cosine itself becomes preference-aware
     │
     └─ DECIDE ──► OUR FIX: never let geometry decide
                   cosine ranks only; LLM returns expresses/opposes/none
```

Orthogonality is the point: **one changes the representation, the other changes the decision rule.** Nothing prevents doing both.

1.3 Comparative performance, read carefully

system	task	score
base ST5-XL	rank: cos(a,match) > cos(a,distractor)?	48.3%
base BGE-large / all-mpnet	same	6.3% / 8.2%
our `text-embedding-3-small`	same	0.5%
DPT-tuned ST5-XL	same	**80.0%**
per-topic rank-20 projection	same	**81.1%**
our LLM judge (isolated)	**decide**: attach match ∧ reject distractor	**95.0%**
our full engine (live corpus)	file into correct synth among ~20 rivals	75.3%

These are different tasks. Theirs is a *ranking* metric — does one similarity exceed another. Ours is a *decision* metric with a threshold: the match must clear confidence ≥ 0.6 *and* the distractor must fail to. Ranking is strictly easier; a system can rank correctly while attaching both or neither. The honest statement is narrow: **on these instances, LLM judgment decides stance more reliably than any geometry tested, tuned or not** — at roughly $10^4\times$ the cost per comparison.

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2. What each method cannot do

	DPT / projection	ELJ
inference cost	**a dot product**	2–3 LLM calls (~$0.2–0.5 / 1k statements)
latency	negligible	1–2 s per statement
demonstrated scale	1.46M pairwise preferences	10^3 statements
training data	synthetic triplets, or per-topic votes	**none**
cold start	projection needs ~50 labelled triplets *for that topic*	works from statement #1
output	continuous metric space	discrete claim graph + counter-edges
downstream	facility location, k-median, proportional representation, ideal points, utilities for unvoted statements	dedup, canonical claims, pro/con structure, synthesis
explanation	a number	a verdict with a stated reason

Two asymmetries drive everything that follows.

Scale. Scoring 1.46M pairwise preferences with an LLM judge is financially absurd. Producing an auditable pro/con claim graph from a projection is impossible — a distance never says "*this opposes that, because…*".

Cold start. Their strongest result (81.1%) requires participant votes **on that topic**. A newly opened question has none. This is exactly the gap the claim registry was built to close, and it is where the loop closes in §3.2.

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3. Four hybrid architectures

3.1 Architecture A — DPT-ranked ELJ (retrieval upgrade)

Change. Replace `text-embedding-3-small` with a DPT-tuned encoder *for ranking only*. The judge, its prompt, and the attach rule are untouched.

Why it should work. Report 1 §7 identifies candidate ranking as the dominant remaining loss. Because distractors outrank matches 98.1% of the time, a cosine-ordered candidate list is **adversarially ordered**:

the top of every list is populated by the statements most likely to be contradictions. Production caps retrieval at the top 10 neighbours (`NEIGHBOR_LIMIT = 10`). At corpus scale, the true match can be pushed out of the window before the judge ever sees it — an error no judge improvement can recover.

DPT targets this exact failure. It moves triplet ranking from 48.3% to 80.0%, i.e. it promotes the true match above the stance-flip — which is precisely what top-K truncation needs.

Predicted effects, with evidence:

effect	mechanism	supporting number
higher recall at fixed K	true match rises in rank	ranking 48.3% → 80.0%
lower cost at fixed recall	shorter lists suffice	judge cost $0.11/1k (short lists) vs $0.41/1k (94-entry lists) — **3.7×**
better cross-language recall	tuned space is stance-, not wording-driven	0% of Hebrew matches reach cosine 0.85 today

Caveat. At pilot scale this is untestable: mean candidate lists were 9.5 synths and 2.7 clusters, so truncation never binds. **Architecture A is a scaling intervention and must be evaluated at 10^3–10^4 statements**, where the top-10 cap becomes active.

3.2 Architecture B — Registry-supervised projection (the closed loop)

This is the most important idea in this report.

Blair et al. close their paper with an open problem:

> *"the gap between the universal tune and the per-topic projected embedding suggests that preferences have both a shared component across topics and a topic-specific component that only voting data on that topic recovers. A natural next step is to generate hard triplets conditioned on a target topic, producing a topic-specific embedding without per-topic votes."*

The ELJ claim registry is a topic-conditioned hard-triplet generator, and it runs anyway. Every judgment it makes is a label:

judge verdict	supervision produced
`expresses` (same synth)	a **positive pair**: same stance, different wording — a *preference match*
`opposes` (counter-edge)	a **hard negative**: same subject, opposite stance — a *semantic distractor*

That is exactly the (anchor, match, distractor) structure DPT trains on. Compare how each side obtains it:

	Blair et al.	ELJ registry
source	2,000 issues (Habermas Machine + Kialo), GPT-4o-mini filter, Claude Sonnet 4 opinion generation, GPT-4o rewriting	**real participant statements**
domain	synthetic, generic political/social	**the actual live question**
cost	a dedicated generation pipeline	**zero — a by-product of filing**
topic conditioning	none (universal tune)	**inherent**

In our 150-triplet pilot the engine produced 218 synths and their counter-edges from 450 statements. A real deliberation of a few thousand statements would yield triplet counts of the same order as their synthetic corpus — **in-domain, on the target topic, with no votes required.**

Their Figure 3 shows the rank-20 projection crossing the universal DPT tune at roughly **50 labelled triplets**. A registry reaches that volume within the first few hundred statements of a single question.

The loop:

```
registry files statements ──▶ emits same–meaning + opposes pairs
          ▲                                  │
          │                                  ▼
shorter, better–ordered           fits a topic–specific rank–20
candidate lists; cheaper,         projection (no votes needed)
higher recall                                │
          └──────────────────────────────────┘
```

Each component supplies the other's missing input. The registry needs a preference-aware ranker and cannot train one; the projection needs topic-specific supervision and cannot generate it.

3.3 Architecture C — Geometry-gated judging (cost control)

Change. Use the tuned geometry to decide *when* to spend a judge call. In an extreme-confidence band, decide geometrically; call the judge only inside the ambiguous margin.

Rationale. Our objection to geometry-as-decision was measured on *base* cosine, where confidence is meaningless (distractor median 0.969 *above* match median 0.871). A preference-tuned space does not share that pathology; Blair et al. report the tuned/projected geometry tracks continuous Likert ratings better than the base model.

Risk — stated plainly. This reintroduces geometry as a decision-maker, which Report 1 argues against. At 80–81% triplet accuracy, ~20% of ranking decisions are still wrong. The defensible version calibrates the bands so that the geometry-only path carries a measured false-merge rate no worse than the judge's 2.1%, and **fails toward the judge** whenever in doubt. If that calibration is not achievable, the architecture should be dropped rather than tuned into plausibility.

Recommendation: evaluate only after A and B; treat as an optimisation, never as an accuracy claim.

3.4 Architecture D — Division of labour across the product

Not a merge but a boundary. The two methods answer genuinely different questions:

question	method	Freedi surface
"is this the same claim?"	ELJ registry	dedup, canonical claims, pro/con, synthesis
"would this participant endorse this?"	preference geometry	deliberation-group formation, slate selection, proportional representation, participation analytics

Blair et al.'s §8 targets are grouping participants, selecting slates, fair representation — social choice, not deduplication. Their own limitations section cautions that the geometry "should not be used as a basis for binding decisions." Neither method should be stretched into the other's question.

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4. Experimental programme

Ordered by information gained per unit cost.

E1 — Rank diagnostic (cheap, no training). For every triplet, record the rank of the true match in the candidate list under (a) `text-embedding-3-small`, (b) an off-the-shelf ST5-XL, (c) the published DPT-tuned ST5-XL. Report rank-of-match distribution and recall@K for K ∈ {1, 5, 10, 20}. *This alone quantifies how much Architecture A can possibly buy, before any integration work.* The DPT model is published (`huggingface.co/cartgr/embeddings-for-preferences-st5-xl`).

E2 — Condition E with a DPT ranker. Re-run the condition E flat-fallback configuration with retrieval swapped to the DPT encoder; judge unchanged. Primary metric: `synthRecall` at fixed candidate budget. Secondary: judge tokens per statement. Powered for the ±5pp noise floor — n = 150 will not resolve small effects; use the full 875.

E3 — Scale test (the decisive one for Architecture A). Build corpora of 10^3–10^4 statements so the top-10 retrieval cap binds. Compare base vs DPT ranking on recall and cost. **Architecture A's value is invisible below this scale**, so E2 alone cannot settle it.

E4 — Registry-as-supervision (Architecture B). From registry output on a single question, extract same-synth positives and counter-edge negatives; fit a rank-20 projection; evaluate on held-out triplets *from that question*. Key comparisons: registry-supervised projection vs vote-supervised projection (81.1%) vs universal DPT (80.0%). Sweep supervision volume to locate the crossover their Figure 3 puts at ~50 triplets.

E5 — Judge-adversarial triplets (see §5). Generate distractors designed to defeat an LLM judge, not an embedding. Re-measure both methods. This is the highest-value experiment for scientific validity, and the least likely to flatter either.

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5. The shared blind spot

The 875 hard triplets were generated by **GPT-4o rewriting anchors to maximise lexical overlap while flipping stance** — an attack engineered against *embeddings*. Nothing in their construction targets a model that reads the text.

This asymmetry cuts against our numbers specifically. Our judge scores 95% against an adversary purpose-built to defeat the component we replaced. Blair et al. have no such asymmetry: their adversary and their solution live in the same representation, so their 80% is measured against a genuinely matched threat.

An LLM-targeted adversary would look different — negation buried in a subordinate clause, scope or quantifier shifts ("some" → "all"), conditional hedges, counterfactual framing, sarcasm, statements that agree on policy but differ on justification. There is no evidence in this work about how either method performs there.

Consequence for the hybrid. A combined system inherits both evaluation gaps. Before deployment claims are made, E5 must run. We consider 95% an upper bound whose tightness is unknown.

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6. Assessment

The complementarity is real, not rhetorical. Each method's principal weakness is the other's output:

weakness	whose	what fixes it
candidate lists ordered adversarially (98.1% inversion)	ELJ	DPT ranking (48.3% → 80.0%)
judge cost scales with list length (3.7× measured)	ELJ	better ranking → shorter lists
best result needs per-topic votes; cold start	theirs	registry-emitted topic-conditioned triplets
no explanation, no claim structure	theirs	ELJ verdicts and counter-edges

Recommended sequence. E1 first — it is nearly free and bounds the upside of the entire programme. If rank-of-match improves materially, proceed to E4 (Architecture B), which is the scientifically novel contribution: a system whose own operation generates the supervision that makes it cheaper, answering an open problem the source paper explicitly poses. E3 decides whether Architecture A matters in production. Architecture C only after A and B, and only if it can be calibrated to a measured false-merge ceiling. E5 governs how strongly any of it may be claimed.

Expected end state. A deliberation platform where a preference-tuned geometry retrieves and orders candidates cheaply at scale, an LLM judge makes every identity decision with a stated reason, and the resulting claim graph feeds back as topic-specific supervision that continuously sharpens the geometry. Neither method reaches that alone: one cannot afford to decide, the other cannot afford to search.